Version 7.4.8

# PVsyst - Simulation report

## Grid-Connected System

Project: 230329 - Hova

Variant: 240423 - Hova - Forecast SRC 5MW

Tracking system with backtracking

System power: 5917 kWp

Källtorp - Sweden

**Author**

Svea Solar Utility Services AB (Sweden)

# Project: 230329 - Hova

## Variant: 240423 - Hova - Forecast SRC 5MW

**Svea Solar Utility Services AB (Sweden)**

## Project summary

**Geographical Site**
**Källtorp**
Sweden

**Situation**

| | |
|---|---|
| Latitude | 58.85 °N |
| Longitude | 14.20 °E |
| Altitude | 109 m |
| Time zone | UTC+1 |

**Weather data**
Källtorp
Solcast TMY

**Monthly albedo values**

| | Jan. | Feb. | Mar. | Apr. | May | June | July | Aug. | Sep. | Oct. | Nov. | Dec. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Albedo | 0.40 | 0.40 | 0.40 | 0.20 | 0.20 | 0.20 | 0.20 | 0.20 | 0.20 | 0.20 | 0.20 | 0.30 |

## System summary

**Grid-Connected System**
Simulation for year no 1

**Tracking system with backtracking**

**PV Field Orientation**
**Orientation**

| | |
|---|---|
| Tracking plane, tilted axis | |
| Avg axis tilt | -0.5 ° |
| Avg axis azim. | 0 ° |

**Tracking algorithm**

| | |
|---|---|
| Irradiance optimization | |
| Backtracking activated | |
| Wind stow | |
| Wind Speed threshold | 15.6 m/s |
| Wind stow position | 0 ° |

**Near Shadings**

| | |
|---|---|
| According to strings : Slow (simul.) | |
| Electrical effect | 70 % |
| Diffuse shading | Automatic |

**System information**
**PV Array**

| | |
|---|---|
| Nb. of modules | 8680 units |
| Pnom total | 5917 kWp |

**Inverters**

| | |
|---|---|
| Nb. of units | 17 units |
| Pnom total | 5312 kWac |
| Grid power limit | 5000 kWac |
| Grid lim. Pnom ratio | 1.183 |

**User's needs**
Unlimited load (grid)

## Results summary

| | | | | | |
|---|---|---|---|---|---|
| Produced Energy | 7110511 kWh/year | Specific production | 1202 kWh/kWp/year | Perf. Ratio PR | 89.27 % |
| Apparent energy | 7110511 kVAh/year | | | | |

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## General parameters

### Grid-Connected System

### Tracking system with backtracking

### PV Field Orientation

**Orientation**

| | |
|---|---|
| Tracking plane, tilted axis | |
| Avg axis tilt | -0.5 ° |
| Avg axis azim. | 0 ° |

**Tracking algorithm**

| | |
|---|---|
| Irradiance optimization | |
| Backtracking activated | |
| Wind stow | |
| Wind Speed threshold | 15.6 m/s |
| Wind stow position | 0 ° |

**Backtracking array**

| | |
|---|---|
| Nb. of trackers | 96 units |

**Sizes**

| | |
|---|---|
| Tracker Spacing | 16.7 m |
| Collector width | 4.78 m |
| Ground Cov. Ratio (GCR) | 28.6 % |
| Phi min / max. | -/+ 60.0 ° |

**Backtracking strategy**

| | |
|---|---|
| Phi limits for BT | -/+ 73.3 ° |
| Backtracking pitch | 16.7 m |
| Backtracking width | 4.78 m |

**Models used**

| | |
|---|---|
| Transposition | Perez |
| Diffuse | Imported |
| Circumsolar | separate |

### Horizon

| | |
|---|---|
| Average Height | 1.4 ° |

### Near Shadings

| | |
|---|---|
| According to strings : Slow (simul.) | |
| Electrical effect | 70 % |
| Diffuse shading | Automatic |

### User's needs

Unlimited load (grid)

### Bifacial system

| | |
|---|---|
| Model | 2D Calculation unlimited trackers |

**Bifacial model geometry**

| | |
|---|---|
| Tracker Spacing | 16.70 m |
| Tracker width | 4.78 m |
| GCR | 28.6 % |
| Axis height above ground | 2.57 m |

**Bifacial model definitions**

| | |
|---|---|
| Ground albedo average | 0.26 |
| Bifaciality factor | 81 % |
| Rear shading factor | 3.0 % |
| Rear mismatch loss | 10.0 % |
| Shed transparent fraction | 0.0 % |

**Monthly ground albedo values**

| Jan. | Feb. | Mar. | Apr. | May | June | July | Aug. | Sep. | Oct. | Nov. | Dec. | Year |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0.40 | 0.40 | 0.40 | 0.20 | 0.20 | 0.20 | 0.20 | 0.20 | 0.20 | 0.20 | 0.20 | 0.30 | 0.26 |

### Grid injection point

**Grid power limitation**

| | |
|---|---|
| Active power | 5000 kWac |
| Pnom ratio | 1.183 |

**Power factor**

| | |
|---|---|
| Cos(phi) (lagging) | 1.000 |

## PV Array Characteristics

### PV module

| | |
|---|---|
| Manufacturer | Trina Solar |
| Model | TSM-685NEG21C.20-FREZNEL |
| (Custom parameters definition) | |
| Unit Nom. Power | 685 Wp |
| Number of PV modules | 2912 units |
| Nominal (STC) | 1995 kWp |

### Inverter

| | |
|---|---|
| Manufacturer | Huawei Technologies |
| Model | SUN2000-330KTL-H1-Preliminary V0.2 |
| (Custom parameters definition) | |
| Unit Nom. Power | 313 kWac |
| Number of inverters | 5.3 units |
| Total power | 1662 kWac |

## PV Array Characteristics

### Array #1 - String_20_685

| | | | |
|---|---|---|---|
| Number of PV modules | 1680 units | Number of inverters | 18 * MPPT 17% 3 units |
| Nominal (STC) | 1151 kWp | Total power | 937 kWac |
| Modules | 60 string x 28 In series | | |
| **At operating cond. (50°C)** | | Operating voltage | 500-1500 V |
| Pmpp | 1064 kWp | Max. power (=>30°C) | 330 kWac |
| U mpp | 1018 V | Pnom ratio (DC:AC) | 1.23 |
| I mpp | 1045 A | | |

### Array #2 - String_19_685

| | | | |
|---|---|---|---|
| Number of PV modules | 1064 units | Number of inverters | 12 * MPPT 17% 2 units |
| Nominal (STC) | 729 kWp | Total power | 625 kWac |
| Modules | 38 string x 28 In series | | |
| **At operating cond. (50°C)** | | Operating voltage | 500-1500 V |
| Pmpp | 674 kWp | Max. power (=>30°C) | 330 kWac |
| U mpp | 1018 V | Pnom ratio (DC:AC) | 1.17 |
| I mpp | 662 A | | |

### Array #3 - String_6_685_Inv1-6

| | | | |
|---|---|---|---|
| Number of PV modules | 168 units | Number of inverters | 2 * MPPT 16% 0.3 unit |
| Nominal (STC) | 115 kWp | Total power | 99.2 kWac |
| Modules | 6 string x 28 In series | | |
| **At operating cond. (50°C)** | | Operating voltage | 500-1500 V |
| Pmpp | 106 kWp | Max. power (=>30°C) | 330 kWac |
| U mpp | 1018 V | Pnom ratio (DC:AC) | 1.16 |
| I mpp | 104 A | | |

| **PV module** | | **Inverter** | |
|---|---|---|---|
| Manufacturer | Trina Solar | Manufacturer | Huawei Technologies |
| Model | TSM-680NEG21C.20-FREZNEL | Model | SUN2000-330KTL-H1-Preliminary V0.2 |
| (Custom parameters definition) | | (Custom parameters definition) | |
| Unit Nom. Power | 680 Wp | Unit Nom. Power | 313 kWac |
| Number of PV modules | 5768 units | Number of inverters | 11.7 units |
| Nominal (STC) | 3922 kWp | Total power | 3651 kWac |

### Array #4 - String_19_680

| | | | |
|---|---|---|---|
| Number of PV modules | 1064 units | Number of inverters | 12 * MPPT 17% 2 units |
| Nominal (STC) | 724 kWp | Total power | 625 kWac |
| Modules | 38 string x 28 In series | | |
| **At operating cond. (50°C)** | | Operating voltage | 500-1500 V |
| Pmpp | 669 kWp | Max. power (=>30°C) | 330 kWac |
| U mpp | 1014 V | Pnom ratio (DC:AC) | 1.16 |
| I mpp | 659 A | | |

### Array #5 - String_18_680

| | | | |
|---|---|---|---|
| Number of PV modules | 1008 units | Number of inverters | 12 * MPPT 17% 2 units |
| Nominal (STC) | 685 kWp | Total power | 625 kWac |
| Modules | 36 string x 28 In series | | |
| **At operating cond. (50°C)** | | Operating voltage | 500-1500 V |
| Pmpp | 633 kWp | Max. power (=>30°C) | 330 kWac |
| U mpp | 1014 V | Pnom ratio (DC:AC) | 1.10 |
| I mpp | 625 A | | |

## PV Array Characteristics

### Array #6 - String_17_680

| | | | |
|---|---|---|---|
| Number of PV modules | 3332 units | Number of inverters | 42 * MPPT 17% 7 units |
| Nominal (STC) | 2266 kWp | Total power | 2187 kWac |
| Modules | 119 string x 28 In series | | |
| **At operating cond. (50°C)** | | Operating voltage | 500-1500 V |
| Pmpp | 2094 kWp | Max. power (=>30°C) | 330 kWac |
| U mpp | 1014 V | Pnom ratio (DC:AC) | 1.04 |
| I mpp | 2065 A | | |

### Array #7 - String_13_680_Inv1-6

| | | | |
|---|---|---|---|
| Number of PV modules | 364 units | Number of inverters | 4 * MPPT 17% 0.7 unit |
| Nominal (STC) | 248 kWp | Total power | 213 kWac |
| Modules | 13 string x 28 In series | | |
| **At operating cond. (50°C)** | | Operating voltage | 500-1500 V |
| Pmpp | 229 kWp | Max. power (=>30°C) | 330 kWac |
| U mpp | 1014 V | Pnom ratio (DC:AC) | 1.16 |
| I mpp | 226 A | | |

### Total PV power

| | |
|---|---|
| Nominal (STC) | 5917 kWp |
| Total | 8680 modules |
| Module area | 26963 m² |

### Total inverter power

| | |
|---|---|
| Total power | 5312 kWac |
| Nb. of inverters | 17 units |
| | 0.0 unused |
| Pnom ratio | 1.11 |

Power sharing defined

## Array losses

### Array Soiling Losses

Average loss Fraction 2.7 %

| Jan. | Feb. | Mar. | Apr. | May | June | July | Aug. | Sep. | Oct. | Nov. | Dec. |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 8.0% | 4.0% | 1.0% | 1.0% | 1.0% | 1.0% | 1.0% | 1.0% | 1.0% | 1.0% | 2.0% | 10.0% |

### Thermal Loss factor

Module temperature according to irradiance

| | |
|---|---|
| Uc (const) | 29.0 W/m²K |
| Uv (wind) | 2.0 W/m²K/m/s |

### Serie Diode Loss

| | |
|---|---|
| Voltage drop | 0.7 V |
| Loss Fraction | 0.1 % at STC |

### LID - Light Induced Degradation

Loss Fraction 1.5 %

### Module Quality Loss

Loss Fraction -0.8 %

### Module mismatch losses

Loss Fraction 2.0 % at MPP

### Strings Mismatch loss

Loss Fraction 0.1 %

### Module average degradation

| | |
|---|---|
| Year no | 1 |
| Loss factor | 0.4 %/year |

**Mismatch due to degradation**

| | |
|---|---|
| Imp RMS dispersion | 0.4 %/year |
| Vmp RMS dispersion | 0.4 %/year |

### IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

| 0° | 30° | 50° | 60° | 70° | 75° | 80° | 85° | 90° |
|---|---|---|---|---|---|---|---|---|
| 1.000 | 0.999 | 0.987 | 0.962 | 0.892 | 0.816 | 0.681 | 0.440 | 0.000 |

## DC wiring losses

| | |
|---|---|
| Global wiring resistance | 0.81 mΩ |
| Loss Fraction | 0.4 % at STC |

**Array #1 - String_20_685**

| | |
|---|---|
| Global array res. | 4.2 mΩ |
| Loss Fraction | 0.4 % at STC |

**Array #2 - String_19_685**

| | |
|---|---|
| Global array res. | 6.6 mΩ |
| Loss Fraction | 0.4 % at STC |

**Array #3 - String_6_685_Inv1-6**

| | |
|---|---|
| Global array res. | 42 mΩ |
| Loss Fraction | 0.4 % at STC |

**Array #4 - String_19_680**

| | |
|---|---|
| Global array res. | 6.6 mΩ |
| Loss Fraction | 0.4 % at STC |

**Array #5 - String_18_680**

| | |
|---|---|
| Global array res. | 7.0 mΩ |
| Loss Fraction | 0.4 % at STC |

**Array #6 - String_17_680**

| | |
|---|---|
| Global array res. | 2.1 mΩ |
| Loss Fraction | 0.4 % at STC |

**Array #7 - String_13_680_Inv1-6**

| | |
|---|---|
| Global array res. | 19 mΩ |
| Loss Fraction | 0.4 % at STC |

## System losses

**Auxiliaries loss**

| | |
|---|---|
| constant (fans) | 30.0 kW |
| 0.0 kW from Power thresh. | |
| Proportional to Power | 7.5 W/kW |
| 0.0 kW from Power thresh. | |

## AC wiring losses

**Inv. output line up to MV transfo**

| | |
|---|---|
| Inverter voltage | 800 Vac tri |
| Loss Fraction | 1.09 % at STC |

**Inverter: SUN2000-330KTL-H1-Preliminary V0.2**

| | |
|---|---|
| Wire section (17 Inv.) | Alu 17 x 3 x 240 mm² |
| Average wires length | 155 m |

**MV line up to Injection**

| | |
|---|---|
| MV Voltage | 12 kV |
| Average each inverter | |
| Wires | Alu 3 x 95 mm² |
| Length | 470 m |
| Loss Fraction | 0.31 % at STC |

## AC losses in transformers

**MV transfo**

| | |
|---|---|
| Medium voltage | 12 kV |

**Transformer from Datasheets**

| | |
|---|---|
| Nominal power | 2500 kVA |
| Iron Loss | 1.58 kVA |
| Iron loss fraction | 0.06 % of PNom |
| Copper loss | 18.50 kVA |
| Copper loss fraction | 0.74 % at PNom |
| Coils equivalent resistance | 3 x 1.89 mΩ |

**Operating losses at STC (full system)**

| | |
|---|---|
| Nb. identical MV transfos | 2 |
| Nominal power at STC | 5.82 MVA |
| Iron loss | 3.15 kVA |
| Iron loss fraction | 0.05 % at STC |
| Copper loss | 50.15 kVA |
| Copper loss fraction | 0.86 % at STC |

## Horizon definition

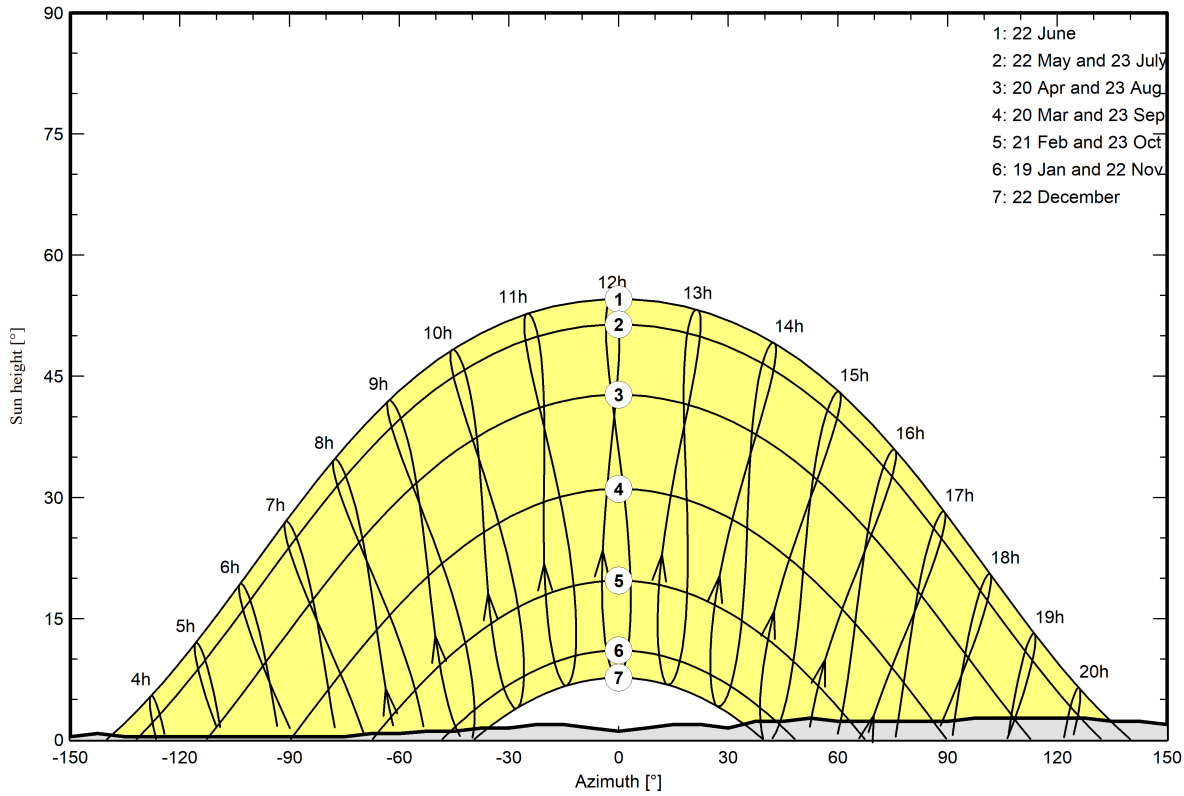
**Horizon from PVGIS website API, Lat=58°51"15', Long=14°11"50', Alt=103m**

| | | | |
|---|---|---|---|
| Average Height | 1.4 ° | Albedo Factor | 0.88 |
| Diffuse Factor | 0.96 | Albedo Fraction | 100 % |

### Horizon profile

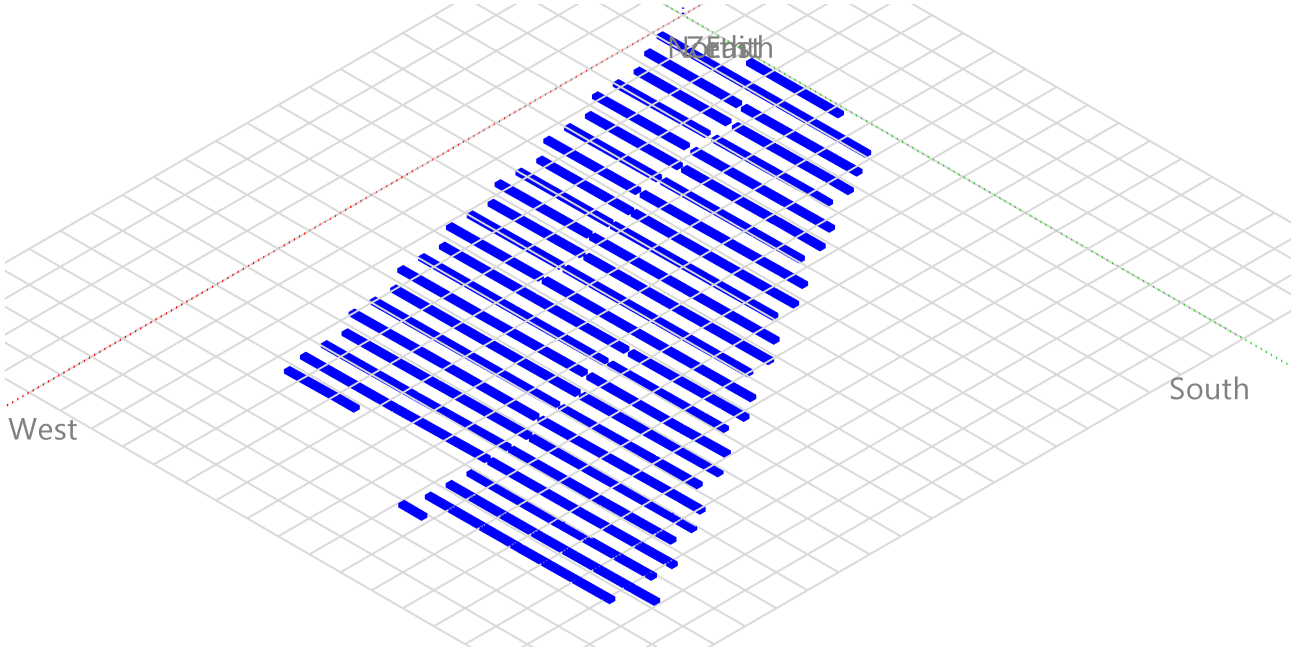
| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Azimuth [°] | -180 | -173 | -165 | -158 | -143 | -135 | -75 | -68 | -60 | -53 | -45 | -38 |
| Height [°] | 0.0 | 0.4 | 0.0 | 0.0 | 0.8 | 0.4 | 0.4 | 0.8 | 0.8 | 1.1 | 1.1 | 1.5 |
| Azimuth [°] | -30 | -23 | -15 | -8 | 0 | 8 | 15 | 23 | 30 | 38 | 45 | 53 |
| Height [°] | 1.5 | 1.9 | 1.9 | 1.5 | 1.1 | 1.5 | 1.9 | 1.9 | 1.5 | 2.3 | 2.3 | 2.7 |
| Azimuth [°] | 60 | 90 | 98 | 128 | 135 | 143 | 150 | 158 | 173 | 180 | | |
| Height [°] | 2.3 | 2.3 | 2.7 | 2.7 | 2.3 | 2.3 | 1.9 | 1.5 | 0.0 | 0.0 | | |

### Sun Paths (Height / Azimuth diagram)



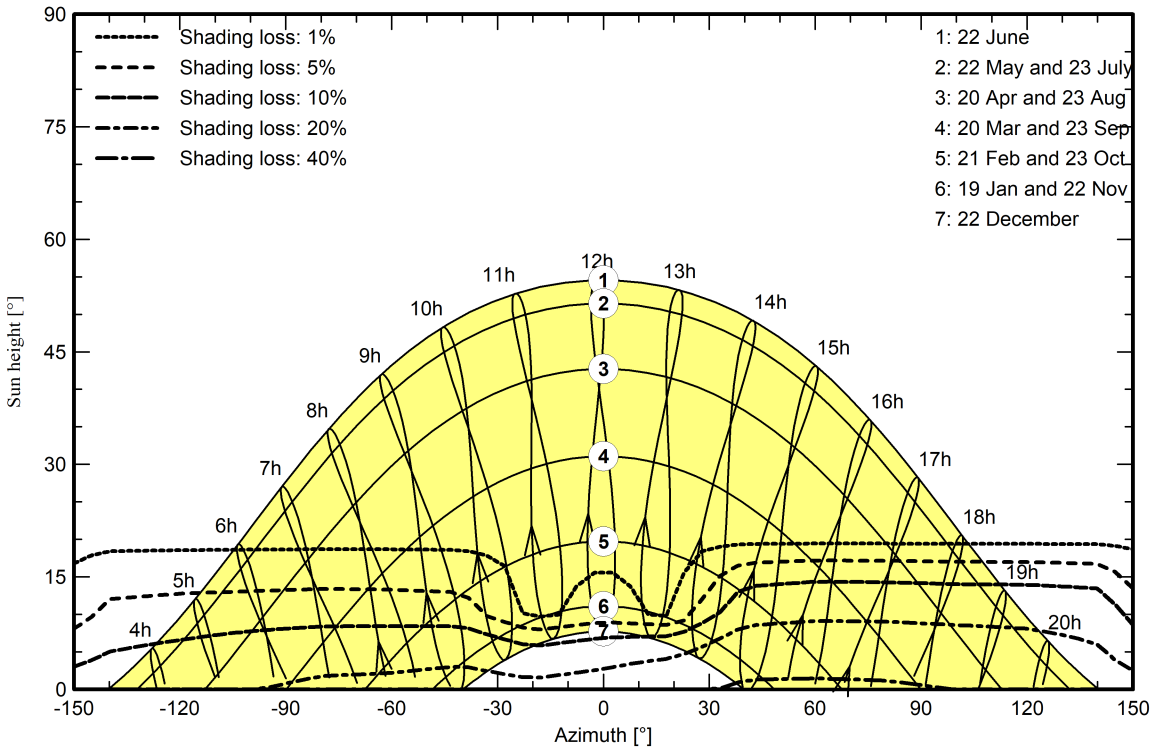
## Near shadings parameter

**Perspective of the PV-field and surrounding shading scene**




## Iso-shadings diagram

**Orientation #1**

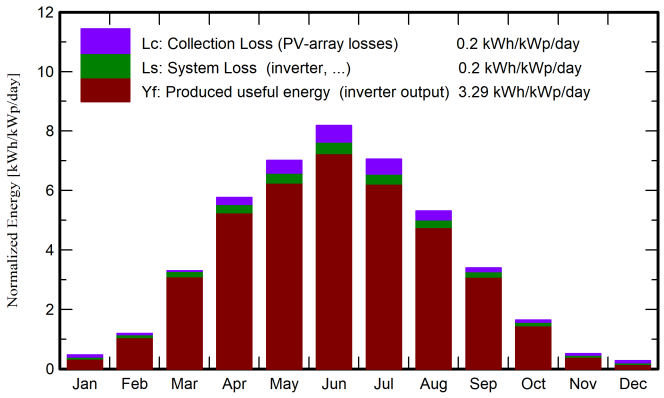


## Main results

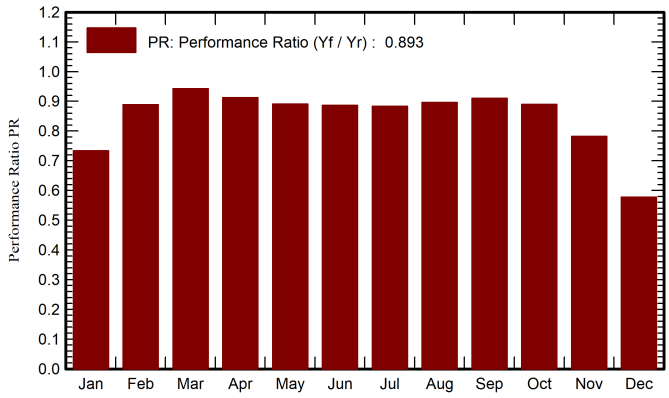
### System Production

| | | | |
|---|---|---|---|
| Produced Energy | 7110511 kWh/year | Specific production | 1202 kWh/kWp/year |
| Apparent energy | 7110511 kVAh/year | Perf. Ratio PR | 89.27 % |

### Normalized productions (per installed kWp)




### Performance Ratio PR



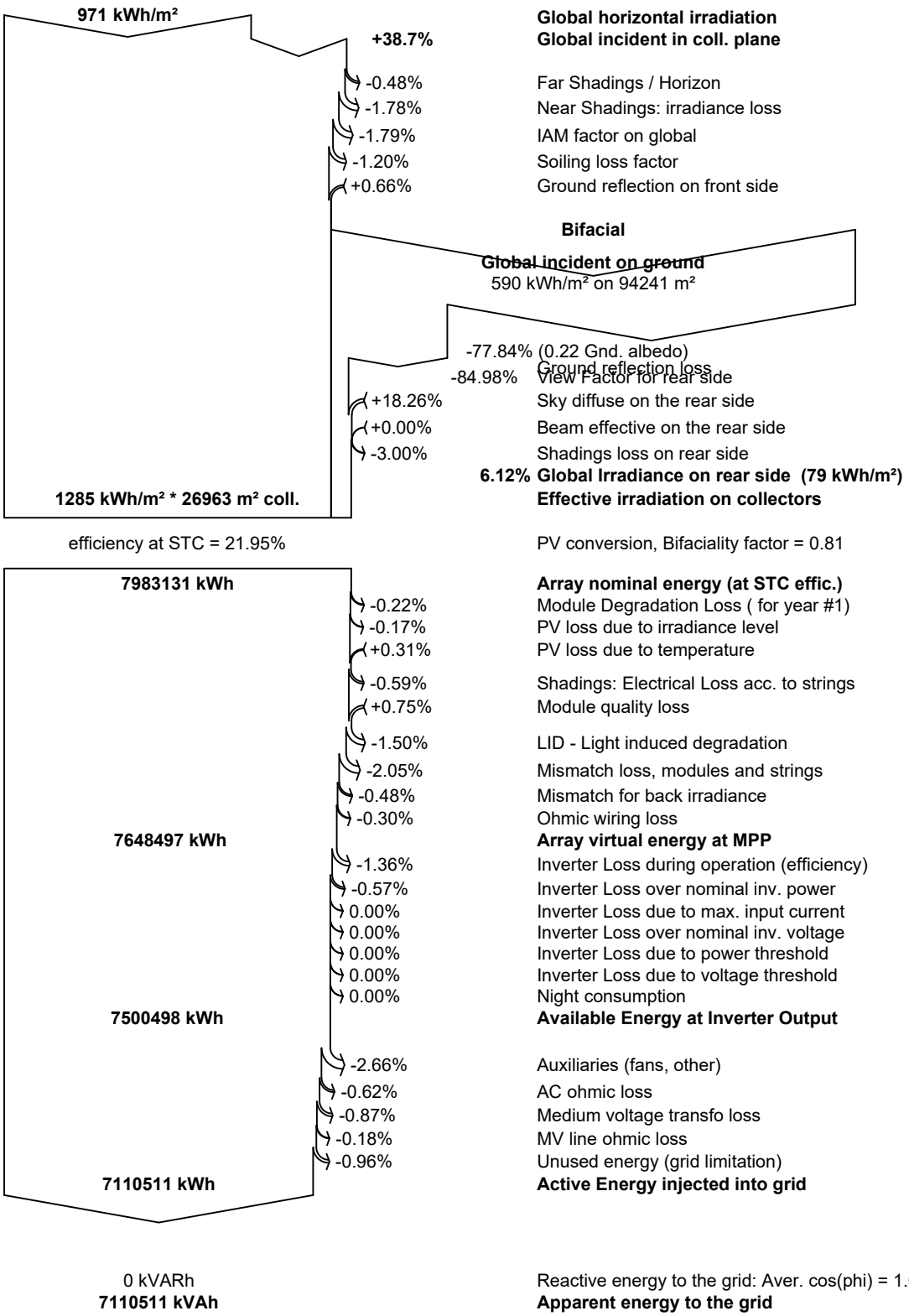

### Balances and main results

| | GlobHor | DiffHor | T_Amb | GlobInc | GlobEff | EArray | E_Grid | PR |
|---|---|---|---|---|---|---|---|---|
| | kWh/m² | kWh/m² | °C | kWh/m² | kWh/m² | kWh | kWh | ratio |
| **January** | 10.4 | 7.02 | 0.80 | 14.4 | 11.5 | 74680 | 62550 | 0.733 |
| **February** | 23.9 | 13.91 | 1.61 | 33.5 | 29.9 | 191868 | 175848 | 0.888 |
| **March** | 69.7 | 28.77 | 3.55 | 102.2 | 96.8 | 602994 | 569748 | 0.942 |
| **April** | 122.1 | 45.48 | 4.43 | 172.9 | 166.3 | 982793 | 933245 | 0.912 |
| **May** | 155.4 | 61.84 | 10.59 | 217.6 | 210.2 | 1208975 | 1147464 | 0.891 |
| **June** | 176.7 | 68.68 | 13.21 | 245.6 | 237.2 | 1356018 | 1287917 | 0.886 |
| **July** | 160.5 | 69.16 | 17.37 | 218.7 | 211.0 | 1204468 | 1143080 | 0.883 |
| **August** | 121.7 | 63.26 | 15.42 | 164.7 | 158.2 | 921662 | 873457 | 0.896 |
| **September** | 74.6 | 41.81 | 11.30 | 102.0 | 97.0 | 581730 | 549343 | 0.910 |
| **October** | 37.3 | 24.24 | 6.89 | 50.8 | 47.3 | 288603 | 267636 | 0.890 |
| **November** | 12.0 | 9.46 | 3.94 | 15.4 | 13.6 | 83577 | 71092 | 0.782 |
| **December** | 6.3 | 4.67 | 2.50 | 8.5 | 6.4 | 39243 | 29132 | 0.578 |
| **Year** | 970.7 | 438.31 | 7.68 | 1346.1 | 1285.3 | 7536611 | 7110511 | 0.893 |

### Legends

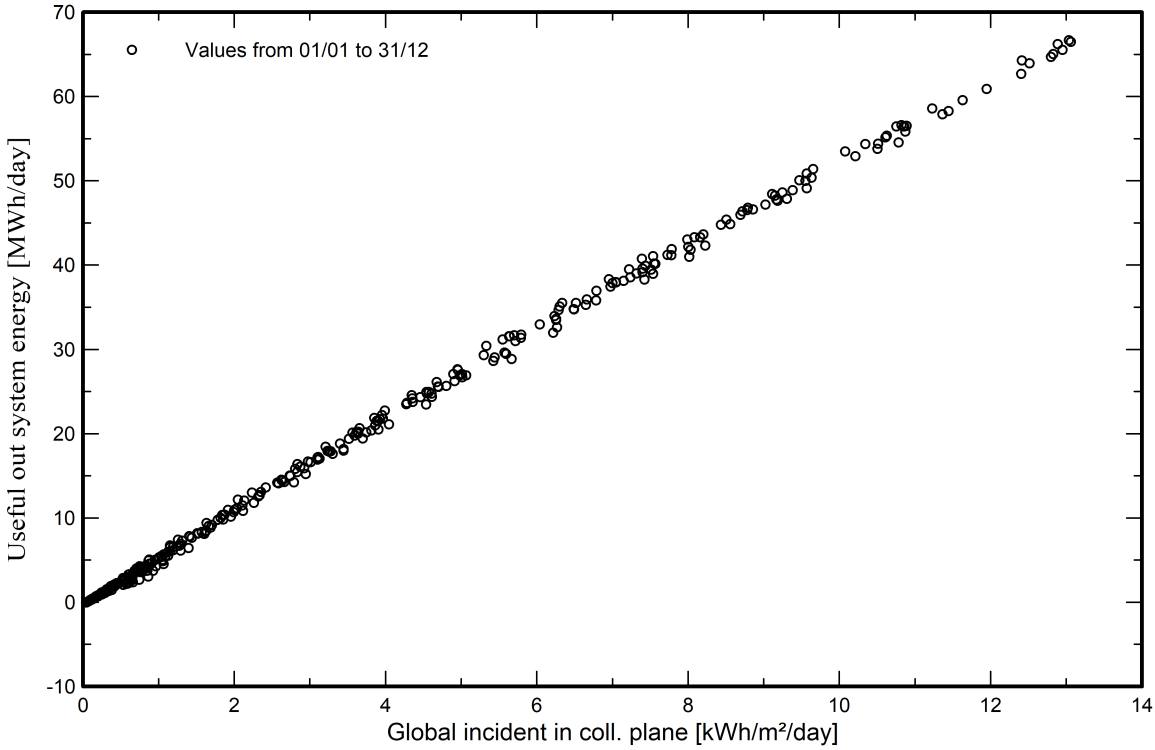
| | | | |
|---|---|---|---|
| GlobHor | Global horizontal irradiation | EArray | Effective energy at the output of the array |
| DiffHor | Horizontal diffuse irradiation | E_Grid | Energy injected into grid |
| T_Amb | Ambient Temperature | PR | Performance Ratio |
| GlobInc | Global incident in coll. plane | | |
| GlobEff | Effective Global, corr. for IAM and shadings | | |

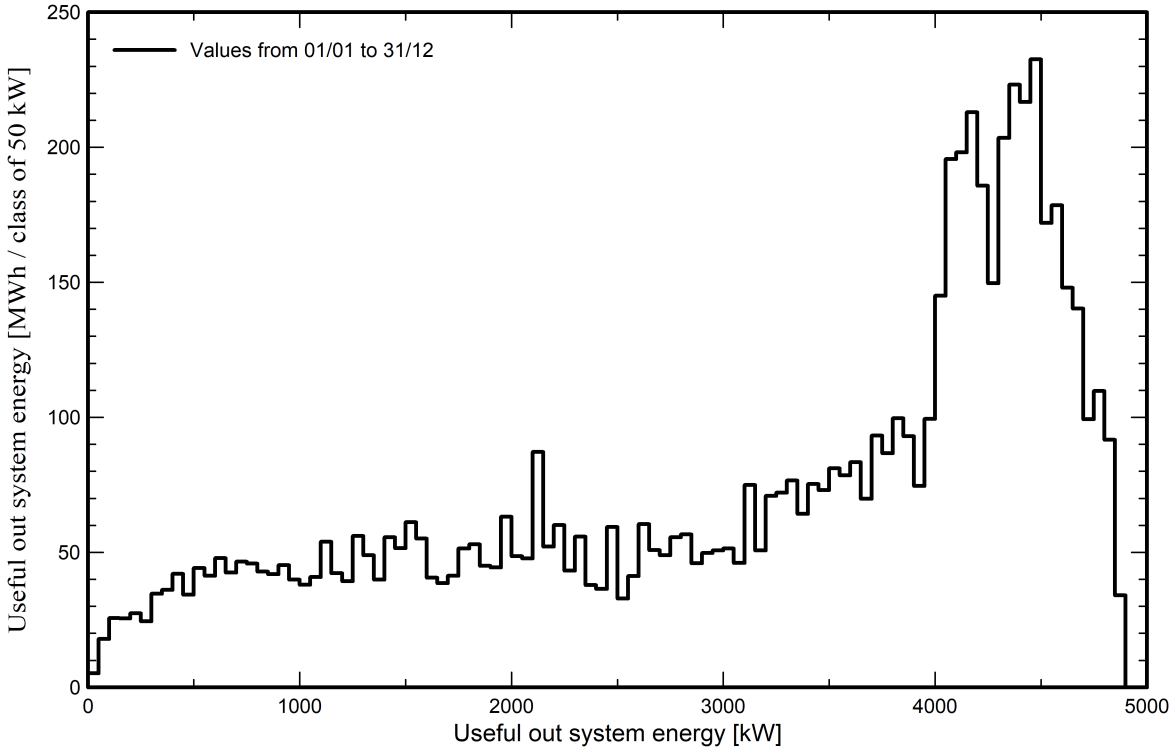
## Loss diagram



## Predef. graphs

### Daily Input/Output diagram




### System Output Power Distribution